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A simulation model of alluvial stratigraphy

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ABSTRACT

The quantitative model presented simulates the development of a two-dimensional alluvial sedimentary succession beneath a floodplain traversed by a single major river. Several inter-related effects which influence the distribution of channel-belt sand and gravel bodies within overbank fines are accounted for. These are (a) laterally variable aggradation, (b) compaction of fine sediment, (c) tectonic movement at floodplain margins, and (d) channel avulsion. Selected experiments with the model show how the interconnectedness and areal density of channel-belt deposits decrease with increasing floodplain width/channel-belt size, mean avulsion period, and channel-belt aggradation rate. Separation of stream patterns based on interconnectedness and channel deposit density is difficult. Tectonic movements do not have a significant influence upon the successions unless a preferred direction of tilting is maintained (half-graben). Then channel-belt deposits showing offlap tendencies tend to cluster adjacent to the active floodplain margin, leaving dominantly fine-grained alluvium to accumulate on the inactive side. Individual channel-belt deposits thicken during aggradation, although a self-regulating limit to such thickening is likely to operate. 'Multistorey' features resulting from aggradation may be difficult to tell apart from those arising through superposition of distinct channel-belt deposits of avulsive origin.

INTRODUCTION

It has long been recognized (reviews in Allen, 1965) that alluvial sedimentary successions are composed of two main facies groups. One group comprises sandstones and conglomerates occurring in sheet or ribbon-like bodies with patterns of texture and internal sedimentary structure indicative of river channel bar deposition. These coarse members may be highly porous and permeable, and define a large group of important hydrocarbon, water and metaliferous reservoirs and ore bodies (e.g. Nanz, 1954; Harms, 1966; Fisher & McGowen, 1969; Sestini, 1972; Bowen, 1975; Campbell, 1976). The other group comprises mudstones, siltstones and thin sandstones with evidence for periodic exposure, soil formation, peat accumulation and shallow sheet-floods, indicative of overbank deposition on alluvial floodplains. These fine members are of low porosity and permeability and frequently define caprocks to hydrocarbon reservoirs.

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The geometry and relative proportion of coarse and fine members in different successions is variable, and has been attributed in a general way to the manner in which an aggrading river channel migrates laterally across its floodplain (e.g. Allen, 1965, figs 35 and 36). Rapid and extensive lateral movement of low sinuosity and braided stream channels, virtually continuously or by discrete jumps, leads to a stratigraphic succession dominated by lenticular or sheet-like coarse members. In contrast meandering streams are more 'stable', migrating within well defined meander belts for long periods. Periodic avulsions lead to a succession containing a larger proportion of fine grained alluvium in which are set ribbon-like meander belt coarse members. The preservation potential of fine-grained alluvium thus depends upon the nature and frequency of lateral channel migration. Allen (1974) elaborated on these 1965 models by considering, in a qualitative way, the effects on alluvial stratigraphy of degradation, climate and base-level changes, and specific modes of channel migration.

In a more refined model, Leeder (1978) recently showed quantitatively that the area ratio of coarse to fine members in alluvial sequences, deposited under 'steady-state' conditions of base level and climate, depends upon vertical floodplain aggradation rate (balancing subsidence) and ratio of floodplain width to channel-belt sand body width in addition to the avulsion frequency variable implicit in Allen's models. An attempt was also made to estimate theoretically the newly defined and economically important 'interconnectedness ratio' of coarse members, that is the proportion of individual channel belt sand bodies that touch each other in a given succession. Simulated alluvial successions were based on random models for avulsion periodicity and channel relocation. An independent approach by Allen (1978) follows similar lines but, in addition, attempts to account for the effects on avulsion of the floodplain relief created by buried and contemporary sand bodies.

Both these recent models ignore the effects of compaction, vertical tectonic movements and lateral variation in floodplain deposition rate on floodplain relief and on the potential location of avulsing channels. These important effects and their inter-relationships are too complex to model manually. In this paper we attempt to analyse these processes and explore the effects of the relevant variables on alluvial stratigraphy using a comprehensive computer simulation model.

FLOODPLAIN GEOMETRY

The floodplain under consideration (Fig. 1) has a finite width bounded by obstacles to river migration, such as terraces or fault scarps. It is traversed by a single river which is free to migrate laterally within a specified zone of influence, the channel belt.

The width of the channel belt depends on stream pattern and size. For single curved channels it will be approximately equal to maximum amplitude of river bends (see also Allen, 1974), and will include abandoned as well as active channel curves. For multiple channel (braided) streams the zone of channel influence is the maximum width confined at bankfull stage. Channel migration for any channel pattern (by bank erosion and bar construction) will be assumed to proceed within its channel belt without changing the predefined width. Thus progressive channel migration (sweeping) in a preferred direction, causing channel-belt width to increase in time, will not be

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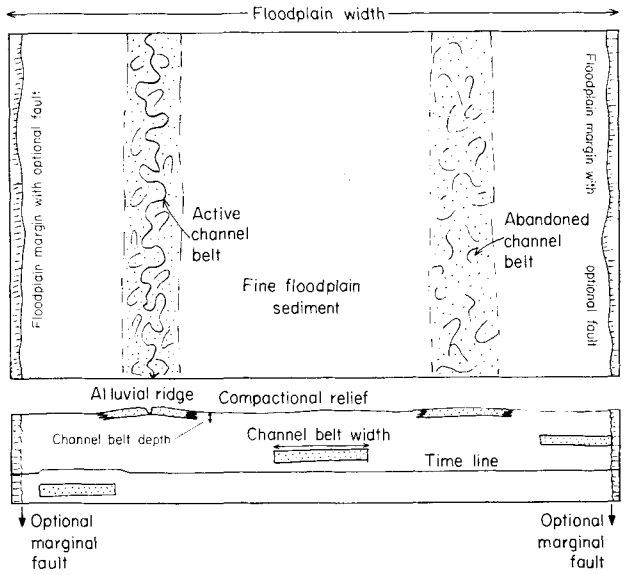

Fig. 1. Definition diagram to show some of the parameters used in the computer model.

considered unless such movements can be looked upon as discrete avulsions separated by short time intervals (e.g. Kosi River; Gole & Chitale, 1966). Floodplain width in natural rivers is often of the order of ten times channel belt width (Fisk, 1947; Russell, 1954; Bernard *et al.*, 1970).

The initial thickness of the channel belt sediments is considered constant across its width and equal to the bankfull depth of the main channel, thus assuming that the channel(s) migrates across the whole belt and can deposit sediment up to the level of its bank. During net aggradation the thickness of the channel belt deposits will increase, but the bankfull stream depth will remain constant with time. Relationships between channel belt-width and bankfull channel depth can be inferred from field data (e.g. Inglis, 1938; Leopold, Wolman & Miller, 1964; Leeder, 1973).

Alluvial stratigraphy will be generated by the model as a vertical section normal to the mean down-valley direction, and to the channel belt. The floodplain surface is taken as initially plane and horizontal in the line of section. For computational expedience a number of reference verticals are set up across the line of section; net aggradation, compaction and tectonic movements are recorded for each vertical and used to reconstruct the simulated stratigraphic succession across the floodplain.

FLOODPLAIN AGGRADATION

A river will overtop its banks just about every year during floods, and deposition may occur from the sediment-laden water. In an equilibrium state, however, the floodplain surface does not become elevated relative to the general level of the stream bed, and the uppermost channel deposits normally attain the level of the floodplain (Wolman & Leopold, 1957). This is because widespread deposition on floodplains by overbank flows is not usual. By inspection of Table 1 it can be seen that average

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Table 1. Floodplain aggradation rates

River	Time scale of deposition	Average deposition rate (cm year^{-1}) or amount (cm)	Range (cm year^{-1} or cm)	Floodplain environment	Source
Ohio	1937 flood	7·0	0·3–56	Floodplain	Mansfield (1938)
Mississippi	30,000 years	—	0·13–0·14	Floodplain	Fisk (1944)
South Carolina Piedmont rivers	150 years	0·8	—	Floodplain	Happ (1945)
Connecticut	1936 (1938) floods	3·5 (2·2)	—	Floodplain	Jahns (1947)
		25·9 (17·3)	—	River banks	
		20·0 (10·7)	—	Tributary banks	
		9·1 (4·3)	—	Artificial levee leesides	
Missouri	1881 flood	—	122–183	Levees	Cited in
Kansas	1951 flood	3·0	—	Floodplain	Wolman & Leopold (1957)
Buck Run	1450 years	0·065	—	Floodplain	and Leopold *et al.* (1964)
Nile	1000 years	—	0·91–1·22	Floodplain	
Tigris and Euphrates	5000 years	0·2	—	Floodplain	
Cheyenne	30–60 years	—	4·1–6·1	Floodplain	
Farmington	1955 flood	< 1·5	0–94	Floodplain	Wolman & Eiler (1958)
Cimarron	12 years	5·2	—	Floodplain	Schumm & Lichty (1963)
Dry Creek	500 years	0·86	—	Floodplain	Brice (1966)
Upper Dry Creek	33 years	—	0·46–0·55	Floodplain	
Well Canyon	40 years	—	1·5–2·0	Floodplain	
Medicine Creek	22 years	8·3	—	Floodplain	
Medicine Creek	Single flood	—	trace–15·2	Floodplain	
Medicine Creek drainage basin	Summary of above	2·5	1·3–7·6	Floodplain	
Chemung	—	0·46	—	Floodplain	Nelson (1966)
Bijou Creek	1965 flood	61–91	0–366	Floodplain (close to channel)	McKee, Crosby & Berryhill (1967)

Table 1 *continued on page* 621

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Table 1 *continued*

River	Time scale of deposition	Average deposition rate (cm year^{-1}) or amount (cm)	Range (cm year^{-1} or cm)	Floodplain environment	Source
Ob'	1969 flood	—	up to 150	Point bar	Velikanov & Yarnykh (1970)
		—	up to 60	Crevasse splay and Levee	
		—	0·2–3	Floodbasin	
Lower Ohio	1964 flood	up to 0·32	—	Accretionary ridge	Alexander & Prior (1971)
Lower Ohio	1964 flood	46	—	Natural levee	
Lower Ohio	750 years	1·0	—	Natural levee	
Lower Ohio	40 years	—	1·5–1·6	Natural levee	
Lower Ohio	1000 years	0·027	—	Accretionary ridge	
Lower Ohio	1000 years	0·6	—	Accretionary ridge	
Lower Ohio	1000 years	0·19	—	Swale	
Wisconsin valley	6040 years	0·035	—	Floodplain	Knox (1972)
Wisconsin valley	6070 years	0·1	—	Floodplain	
Blockhouse Creek	6000 years	—	0·015–0·038	Floodplain	Knox (1972)
Delaware	6000 years	—	0·014–0·115	Floodplain	Ritter, Kinsey & Kauffman (1973)
Mississippi	1973 flood	86	13–300	Point bar	Kesel *et al.* (1974)
		53	10–84	Natural levee	
		12·5	6–27	Levee back	
		6	4–9	Abandoned channels	
		1·1	0·5–2·5	Backswamp	
Western Run	51 years	1·63	—	Floodplain	Costa (1975)
Little Tallahatchie	8(31) years	—	4·7–6·5 (1·3–2·0)	Natural levee	Ritchie, Hawks & McHenry (1975)
		—	2·8–3·4 (2·6–2·7)	Crevasse splay	
		—	0·9–2·8 (0·7–1·0)	Backswamp and abandoned channels	
Upper Mississippi	20 years	—	2·5–3·5	Artificial backwater lakes and pools	McHenry, Ritchie & Verdon (1976)
Beatton	500 years	—	0·1–6·1	Accretion topography	Nanson (1977)
Bobr and Strzegomka	Holocene	—	0·1–0·5	Floodplain	Teisseyre (1977)

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sediment thicknesses deposited by major seasonal floods are only of the order of millimetres or centimetres, and floodplain erosion (both vertical and lateral) occurs locally (Jahns, 1947; Wolman & Leopold, 1957; Wolman & Eiler, 1958; Sigafoos, 1964; Brice, 1966; Velikanov & Yarnykh, 1970; Alexander & Prior, 1971).

The spatial and temporal distribution of vertical deposition and erosion on floodplain surfaces must depend on the hydraulics of overbank flow, on the quantity and size of locally available sediment, and on pre-existing floodplain relief, including nature of vegetation cover (Wolman & Eiler, 1958). Although the hydraulics of overbank flow are rather complicated (e.g. Sellin, 1964; Toebes & Sooky, 1967; Yen & Overton, 1973; Myers & Elsawy, 1975; Ghosh & Kar, 1975) it seems to hold generally that they are primarily responsible for the decrease in thickness and grain size of deposited sediment away from channels (see Allen, 1965; 1970, p. 138–139, and Table 1). Such deposition gives rise to alluvial ridges comprising channel, levee, crevasse and abandoned channel deposits. Erosion may be expected where flow is accelerated in relatively narrow and/or steep flow sections and where vegetation cover is poor (e.g. Wolman & Eiler, 1958; Velikanov & Yarnykh, 1970; Allen, 1970, p. 138). Deposition may occur preferentially in local depressions, like swales (Wolman & Eiler, 1958; Alexander & Prior, 1971), thus tending eventually to smooth out floodplain relief. This is by no means universal, particularly if flow is maintained within the depression. Nanson (1977) records no difference in thickness of sediment deposited in swales and on adjacent ridges, and there are many other examples of deposition (and erosion) on the highest parts of floodplains away from the contemporaneous alluvial ridge (e.g. Wolman & Eiler, 1958; Jahns, 1947).

Long term erosional (degradational) or depositional (aggradational) trends across floodplains are caused by external factors which effect changes in the water and sediment supplied by the headwaters, or in the valley slope (i.e. 'independent' variables). For example, climatic changes and river diversions can modify the balance between sediment and water discharge. Tectonic movements, sediment compaction or eustatic sea level changes may cause valley slope to vary independently (e.g. Fisk, 1944; Coleman, 1969; Leopold *et al.*, 1964).

The dimensions, shape, slope and pattern of alluvial rivers are delicately adjusted to transport the amount of water and sediment supplied to them (valley slope is another controlling variable, however in the long term this may be changed by progressive river erosion or deposition). When aggradation or degradation results from changes in these 'independent' variables an accompanying change in river characteristics is also expected (Schumm, 1971). A notable example is the change from a braided to a meandering lower Mississippi in the Holocene due to a rise in base level (effectively reducing valley slope), a decrease in water discharge, and a reduction in the size of sediment transported (Fisk, 1947). As glaciations are rare in the geological record it seems unlikely that changes like those in the Mississippi are very common. Very small changes in independent variables can cause appreciable aggradation (Leopold *et al.*, 1964), thus as long as continued aggradation can be accommodated by crustal subsidence it seems likely that (dependent) river characteristics would change so slowly that they could be assumed constant for time scales of thousands of years. The mean height difference between floodplain surface and river bed is also expected to remain essentially constant during aggradation (Wolman & Leopold, 1957).

In the model, aggradation (and not degradation) will be considered without taking

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explicit account of the causative factors. It could be accommodated by a rise in land (and sea) level or by subsidence with land level constant, or a combination of both. Cognizant of the preceding discussion, a physical treatment of floodplain deposition must be ruled out. Instead the mean net deposition rate, r, at any distance from the edge of the channel belt, z, is given empirically as

$$r = a(z+1)^{-b} \tag{1}$$

where a is the maximum net deposition rate at the edge of the channel belt, and b describes the rapidity with which the rate of deposition decreases with distance from the channel belt. Equation 1 is inferred from the empirical equation presented by Kesel *et al.* (1974), but there are three important differences. First, Kesel *et al.*'s equation is based on deposition from a single flood and does not represent mean net deposition rate. Second, Kesel *et al.* refer deposition to distance from the river whereas Eqn 1 refers to distance from the edge of the channel belt. In fact vertical deposition within the channel belt may occur on channel bars, levees, crevasse splays and abandoned channel fills, all of which will change position with time as channels migrate. In our model, aggradation rate across the channel belt section is assumed constant and equal to a. The third point is that the original equation of Kesel *et al.* was based on a regression analysis, and if it had been used in a simulation model the mean values predicted should perhaps have had a stochastic component added in order to account for the variance inherent in the original data. For simplicity, Eqn 1 here is taken as deterministic.

Realistic values of a and b can be inferred from Table 1, bearing in mind that deposition rate over a single flood period is not the same as net deposition rate (see also Leeder, 1975, p. 463). It should also be noted that some of the cited rates of point bar deposition may include a component of lateral deposition.

In our model the floodplain surface is redefined each time the channel belt is abandoned, the amount of deposition being the product of net deposition rate (from Eqn 1) and the time period separating avulsions. No sediment can be deposited above the level of the contemporary channel belt because if sediment was deposited on relatively upstanding areas in the floodplain at least as much would also be deposited in the channel belt. Thus areas standing above a given channel belt level will receive no sediment, nor will they be eroded.

Channel-belt sediment will be considered to be dominantly sand, whereas the remaining floodplain sediment will be considered to be dominantly silt and clay (see Allen, 1965, pp. 130–134, and references in Table 1), although in reality sand content might be expected to increase towards the channel belt.

COMPACTION

All sediments deposited with an initial porosity compact with burial, and compaction will continue as long as the depth of burial increases, until porosity becomes negligible. Tectonic stresses will not be considered here.

The base of a layer of finite thickness must have undergone more compaction than the top of the layer as gravitational stress increases progressively with depth (Perrier

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& Quiblier, 1974). Considering a layer of recently deposited sediment, the solid thickness of an infinitely small thickness of this layer is

$$dT_s = [1 - p(y)]dT \tag{2}$$

where T is layer thickness, $p(y)$ is porosity (a function of depth, y, of the lamina), and T_s is the solid thickness. The total solid thickness of the layer with top at ground level and base at distance T below ground level is

$$T_s = \int_0^T [1 - p(y)]dy \tag{3}$$

Assuming that no mineral matter is added, removed or otherwise transformed in density the solid thickness remains constant irrespective of depth of burial. Thus if the solid thickness of a sediment layer is known, changes in total thickness with burial can be easily obtained using

$$T_s = \int_{y_1}^{y_2} [1 - p(y)]dy = \text{constant} \tag{4}$$

where y_2 and y_1 are the burial depths of the bottom and top of the sediment layer, respectively (see Perrier & Quiblier, 1974).

In order to evaluate Eqns 3 and 4 the porosity-depth function must be known. This varies with grain texture, composition, and many other factors (summaries in Rieke & Chilingarian, 1974; Chilingarian & Wolf, 1975, 1976). For clay-rich sediments, however, Baldwin's (1971) composite curve seems to be a good approximation, and for average channel sands porosity reduction with burial can be assumed negligible, at least for depths up to about 0·5 km (Perrier & Quiblier, 1974; Wolf & Chilingarian, 1976).

To find a convenient algebraic expression for porosity in terms of depth for clay-rich sediments a least-squares polynomial was fitted to data points taken from Baldwin's curve. The regression of porosity on the logarithm of depth (plus unity) was, to two significant figures:

$$p(y) = 0{\cdot}78 - 0{\cdot}043 \ln(y+1) - 0{\cdot}0054[\ln(y+1)]^2 \tag{5}$$

This second degree polynomial explained significantly more variance than the linear fit, but the third degree polynomial was not a significant improvement over the second. After inserting Eqn 5 into either Eqns 3 or 4 an analytical expression for the integral is possible:

$$T_s = 0{\cdot}19y + (y+1)\ln(y+1)[0{\cdot}032 + 0{\cdot}0054 \ln(y+1)] \Big|_{y_1}^{y_2} \tag{6}$$

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Thus, given T_s and the depth of one layer boundary, the depth of the other boundary can be obtained from Eqn 6.

In the model only the non-channel-belt deposits are compacted using Eqn 6, ignoring any internal variability in these sediments. Although in reality floodplain deposition probably occurs annually, the model adds the accumulated sediment to the floodplain surface when avulsion occurs (see previous section). Just before the channel moves, all layers below this newly deposited layer are compacted; the newly deposited layer is not compacted until the next avulsion, but its solid thickness is calculated in readiness.

The depth of the layer immediately below the new deposit, and its solid thickness, is known. The compacted thickness is calculated by solving Eqn 6 iteratively (Newton–Raphson method), providing that the compacting layer is not a channel-belt deposit. The upper boundaries of both the new deposit and this compacted layer are accordingly 'packed down' and the depth to the top of the next layer is thus obtained. On re-application of Eqn 6 the compacted thickness of this next layer is found and all boundaries above its base are again packed down. This procedure is repeated down to the base of the section for each vertical. The base of the section remains artificially at a constant position during this process so that differential compaction below this base-level does not occur.

TECTONIC MOVEMENTS

Apart from compactional subsidence, tectonic movements have an important effect on floodplain surface relief, and hence on river behaviour. Net aggradation and subsidence have approximately kept pace with each other for long periods of geological time, as indicated by large sedimentary thicknesses of essentially similar facies. Although measurements of recent vertical crustal movements are common (e.g. Rikitake, 1976; Pavoni & Green, 1975; Green, 1974) the roles of compaction, elastic crustal warping, and relatively sudden release of elastic strain along rupture zones during the subsidence of the land surface are not always obvious.

Recently Potter (1978) has indicated that many large rivers follow structural lows, particularly rifts or other fault bounded depressions. The strong influence of faulting (sometimes rifts) on floodplain topography and river behaviour has also been reported by Fisk (1944), Russell (1954), Coleman (1969) and Mike (1975). Crustal rupture can therefore play an important role in subsidence. Of particular relevance here is the tilting of floodplain surfaces in directions normal to the downvalley direction.

In the model developed so far it is implicit that progressive aggradation is accommodated by compaction, by changes in base level, or (mainly) by tectonic subsidence, although the relative contributions of the last two factors is not explicitly stated. The tectonic subsidence might be associated with elastic downwarping or with periodic rupture at the edges of the floodplain, or a combination of both. No tilting of the floodplain normal to the mean downvalley direction has been assumed thus far during such tectonic subsidence. An option therefore exists to allow tilting of the floodplain surface in the line of section at certain discrete time intervals associated with crustal rupture at floodplain margins.

The occurrence of crustal ruptures (earthquakes) can be treated as a series of point events (Cox & Lewis, 1966). If the events are completely random in time they must

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follow a Poisson process, and times separating successive events must follow a negative exponential distribution. The probability density function, $f(t)$, and distribution function, $F(t)$, are

$$f(t)=\rho^{-1}e^{-t/\rho} \text{ and } F(t)=1-e^{-t/\rho} \qquad (t>0) \tag{7}$$

where e is base of natural logarithms, ρ is the expected value of t, and t is time between events. Such a model implies that successive events are mutually independent, the chance of two or more events occurring simultaneously is negligible, and that the expected time between events does not change (no trend). Large earthquakes (magnitude >6) and some volcanic eruptions probably follow the Poisson process (Lomnitz, 1974; Reyment, 1969; Rikitake, 1976), and Price (1974, 1976) used a Poisson model to simulate times separating uplifts in an alluvial fan model.

If it is assumed that crustal strain is zero after a large earthquake, and that strain is gradually built up until its release by the next earthquake, then the likelihood of an earthquake must increase progressively with time. The distribution that best fits this situation is the Weibull distribution (Hagiwara, 1974), which has probability density and distribution functions

$$f(t)=\frac{c}{\alpha}\left(\frac{t}{\alpha}\right)^{c-1} e^{-(t/\alpha)^c} \text{ and } F(t)=1-e^{-(t/\alpha)^c} \qquad (c>0,\ \alpha>0). \tag{8}$$

The expected value of t is $\alpha\Gamma[(c+1)/c]$ where Γ denotes a gamma function (see Johnson & Kotz, 1970, for further details).

More refined models for earthquake occurrence are possible if there is evidence of a trend, or a memory effect between successive events; such models might be more generalized renewal processes, e.g. semi-Markov processes (see Reyment, 1969, 1976; Wickman, 1976). Only the simple Poisson and Weibull distributions will be used by us.

The amount of tilting at each event (specifically the relative changes in elevation of the section base at either floodplain margin as defined here) is assumed to follow a normal distribution, but unfortunately no reliable data exist to check this assumption. Positive values will be taken as relative depression of the left hand side of the floodplain, negative values relative uplift of the left hand side. By appropriate choice of a mean value, a preferred direction of tilting could be accounted for. If the mean is taken as zero, however, tilting in either direction would be equally likely and, as individual tails of a normal distribution can be approximated by exponential distributions, the amount of tilting in a given direction may be considered almost random (see Price, 1976; Lomnitz, 1974, p. 95). Movements associated with foreshocks and aftershocks are assumed integrated with the main movement for any event.

If the option for tilting is used in a particular experiment the net effects of tilting are assessed, as with compaction and aggradation, just before avulsion of the channel belt. The total number of tectonic events (however defined) and their associated movements during the time period up to the current avulsion are obtained by Monte Carlo sampling from the appropriate distributions (see Harbaugh & Bonham-Carter, 1970). The section base remains planar during tilting.

This procedure implies that aggradation does not respond to progressive compaction or tilting until the channel belt avulses; however the nature of such continuous

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adjustment of local erosion and deposition to compaction and continuous or discrete tilting movements would likely be very complex. Unfortunately present understanding of alluvial processes would not allow consideration of this aspect.

Guidelines for choice of input to these models are difficult to give as specific data are sparse, however some observations from Rikitake (1976) and Lomnitz (1974) may be useful. Slow vertical crustal movements (discrete or continuous) occur of the order of mm year^{-1} (α-phase of Mescherikov, 1968). These may or may not be associated with faults. If vertical crustal movements exceed 1 cm year^{-1} they are probably associated with earthquakes (and faults). Faults associated with large earthquakes (magnitude >6) can cause relative elevation changes in their vicinity of the order of d*m* or *m*, and ground tilting can often be 10^{-4}–10^{-6}. In tectonically active areas earthquakes with magnitude >6 occur with mean recurrence periods of the order of 10^1–10^3 years, becoming less frequent as magnitude increases. Thus large earthquakes along the sides of one valley will be expected to have longer mean recurrence times (e.g. Machette, 1978, p. 878).

AVULSION

Avulsion is the term used to describe a river's relatively sudden abandonment of a channel belt in favour of a new course. The process occurs with meandering and braided rivers and is recorded by abandoned channel belts preserved on floodplains (e.g. Fisk, 1944, 1947, 1952; Russell, 1954; Coleman, 1969; Anderson, 1961; Bernard *et al.*, 1970; Mike, 1975; Knight, 1975; Chitale, 1977). The periodicity of avulsion appears to be of the order of 10^2–10^3 years, often about 10^3 years. The diversions are actually gradual but can be considered instantaneous compared with the recurrence time of avulsion.

Avulsion is initiated if floodwaters travelling from an existing alluvial ridge to the floodbasin, through crevasses or low parts of levees, have a gradient advantage over the main channel. Sometimes the process may be aided by the river intersecting a formerly abandoned channel segment (Coleman, 1969; Chitale, 1977; Fisk, 1952) or a fault zone (Fisk, 1944). Avulsion is more likely with greater relative elevation of the alluvial ridge and floodbasin and higher and more prolonged seasonal flooding. It appears that tectonic activity is very important in controlling avulsion by virtue of its effect on floodplain topography, and that diversions are often established along lows associated with major faults.

In the model it is necessary to predict the length of time between avulsions occurring upstream of the cross-section defined. The uncertainty concerning the occurrence of avulsions during high water periods in any given physiographic setting strongly suggests a stochastic model. It would seem desirable to define the probability of occurrence of avulsion as a function of at least (1) relative elevation of alluvial ridge and floodbasin, (2) a measure of the severity and persistence of seasonal floods. Both of these factors are time dependent, and, by virtue of the progressive nature of alluvial ridge growth, avulsion would become more likely as time progressed. Our knowledge of avulsion processes is unfortunately very sparse, and the structure of such a stochastic model would be purely speculative (but see Bridge, 1975, for a comparable example applied to cut-off). As a surrogate, avulsion recurrence times will be obtained by Monte Carlo sampling from a presumed distribution. If recurrence times

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are completely random they would follow a Poisson process (i.e. Eqn 7). But as avulsion becomes more likely as time progresses, by analogy with earthquake occurrence (previous section) a Weibull distribution (i.e. Eqn 8) is more suitable. A uniformly distributed random sample (Leeder, 1978) would be unsuitable as the artificial constraint at the time boundaries cannot be justified.

Once the time period separating an avulsion from its predecessor is obtained the appropriate net aggradation, compaction and tectonic activity during this time period are calculated and the floodplain surface is adjusted accordingly. The channel belt is then moved and located in the topographically lowest part of the floodplain. It is therefore assumed that avulsion occurred sufficiently far upstream to enable the channel to have shifted to this most favourable position. If there are a number of equally low positions the one nearest the present channel belt is chosen. Leeder (1978) and Allen (1978) position an avulsing channel randomly on the floodplain surface, although Allen uses arbitrary rules to avoid high areas underlain by sand bodies.

COMPUTATIONAL ASPECTS

The composite model presented has been translated into a FORTRAN IV computer program, full details of which are given elsewhere (Bridge, 1979). The program computes, at discrete time intervals, the geometry of successive floodplain surfaces and channel belt deposits across the predefined section, resulting from the processes of aggradation, compaction and vertical tectonic movements. Each new floodplain surface is defined and all previous ones appropriately redefined immediately prior to an avulsion. The geometry of the new surface defines the location of the new channel belt. A complete record of events is printed out during each run of the program, and after a required number of avulsions, a stratigraphic section is drawn with successive floodplain surfaces (i.e. time lines) and channel belts indicated. A scan of the reference columns in this section is conducted to calculate the areal proportion of channel belt deposits (channel deposit density), a measure of the interconnectedness of channel belt deposits (interconnectedness ratio), and the number of separate channel belt occurrences in each reference vertical. By virtue of the scanning procedure the channel deposit density is slightly overestimated, but only by a few per cent if the number of reference verticals is large. Interconnectedness is the ratio of total number of erosive contacts between channel belts to total number of channel belts encountered in the scan; it is therefore a measure of the length of contact between channel belts. Vertical contacts are not considered here as they are a relatively small fraction of the total length of contact. The ratio varies from 0, when no channel belt deposits are interconnected, to $(n-1)/n$ (n is total number of channel belts) when all channel belts have all of their horizontal boundaries in contact with another. The maximum value of the ratio approaches unity as n increases.

The equations developed by Leeder (1978) and Allen (1978) for channel deposit density do not allow for overlapping channel belts, a point, however, recognized by Allen. Leeder's (1978) interconnectedness ratio does not give an adequate representation of the complex ways in which channel sand bodies may be connected, partly because the probability of more than two-fold connection was not accounted for. Allen (1978) defined three superior measures of connectedness but all must be determined experimentally; his 'average fractional contact' is very similar to the inter-

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connectedness ratio defined here. It seems likely that automated scanning of experimental cross-sections must replace simplistic equations as the full complexity of the system is realized.

EXPERIMENTAL PROCEDURE

There are many variables in the model, and it is not intended to attempt a comprehensive evaluation of the effects of all of these. Rather we will examine certain parameters which are considered to have the most important effect on simulated alluvial stratigraphy. These are (a) ratio of channel belt size to floodplain width, (b) mean avulsion rate, (c) channel-belt aggradation rate, and (d) mean period, magnitude and sense of successive tectonic events. In order to facilitate comparisons between simulation experiments a stereotype experiment has been set up with initial variables shown in Table 2. One or more of the above listed parameters is/are then given various different values in order to examine the effects of their variation or

Table 2. Input data for experiments

Stereotype experiment		
Floodplain width	10,000 m	
Channel belt width	1,000 m	
Channel bankfull depth	5 m	
Initial position of channel belt		Centre of floodplain
Distance between reference verticals		100 m
Mean channel-belt aggradation rate		0·02 m year^{-1}
Exponent in aggradation rate model (*b*)		0·1
Mean avulsion period		890 years
Exponent in avulsion model (*c*)		3·3
Tectonics parameters		None
No. of avulsions simulated		50
Set 1 experiments		
Channel belt widths and bankfull depths of 600, 1000, 2000 m and 3, 5, 7 m, respectively.		
Set 2 experiments		
Mean channel-belt aggradation rates of 0·005, 0·01, 0·02, 0·04 m year^{-1}, each at mean avulsion periods of 111, 222, 445, 890, and 1780 years.		
Set 3 (full graben) experiments		
Mean period between tectonic events of 89, 445, 890, 4450 years, and standard deviation of relative vertical movement at floodplain margins of 0·5 and 1·0 m for each period. Mean vertical movement is zero, and exponent *c* in tectonics model is 2·0.		
Set 4 (half graben) experiments		
Mean period between tectonic events of 89, 445, 890, 4450 years, and mean relative vertical movement at floodplain margins of 0·5 and 1·0 m (standard deviations 0·25 and 0·5 m respectively) for each period. Exponent *c* is 2·0.		

co-variation. The experiments are conveniently grouped into four sets as shown in Table 2; input follows the guidelines given in earlier parts of this paper so that nature is not violated. The results of the experiments are illustrated and discussed below.

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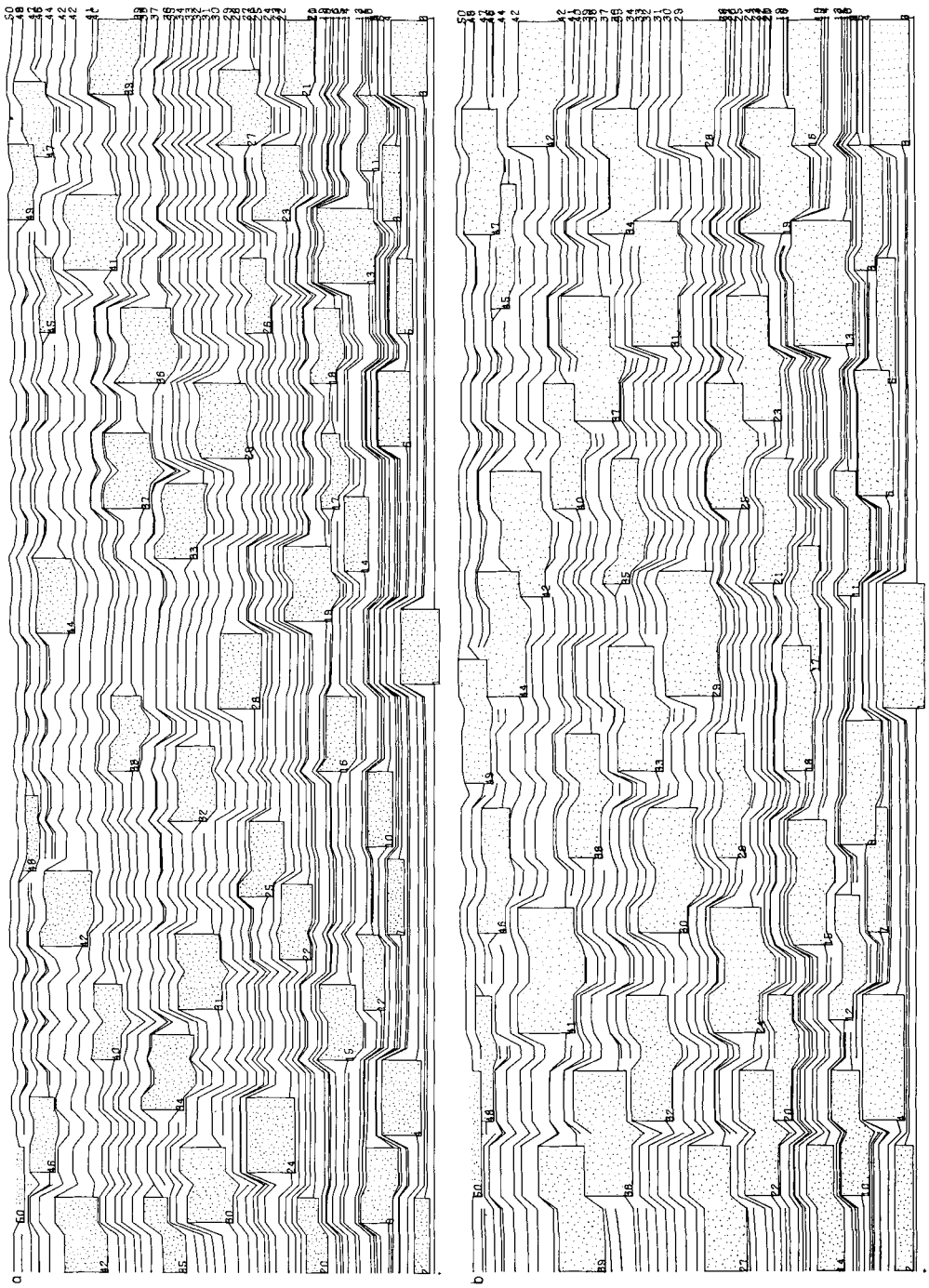

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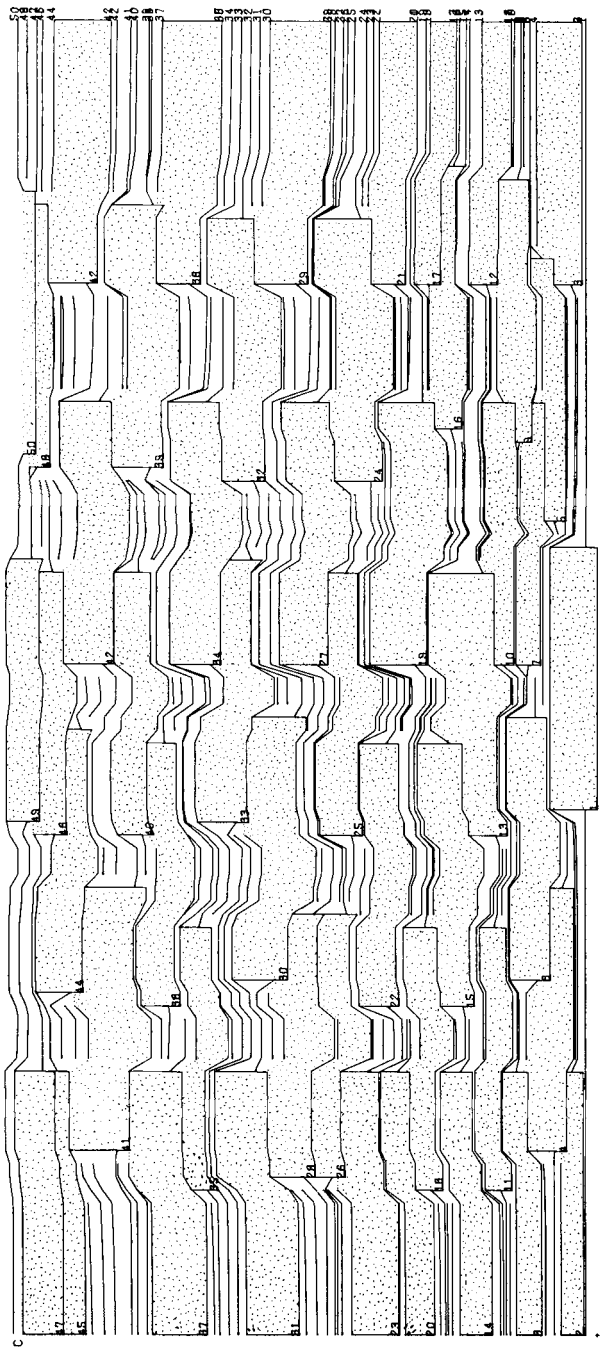

Fig. 2. Simulated cross-sections for set 1 experiments. Vertical exaggeration is 12·5. Channel-belt deposits (stippled) and floodplain surfaces existing at each avulsion are numbered consecutively; floodplain surfaces that have been reworked by a later channel-belt stop slightly short of the eroding channel-belt margin. Input data for all experiments are given on Table 2. Channel belts widths and depths are (a) 600 and 3 m, (b) 1000 and 5 m, (c) 2000 and 7 m.

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CHANNEL DEPOSIT DENSITY AND INTERCONNECTEDNESS

For set 1 experiments, Figs 2 and 3 show channel deposit density and interconnectedness to increase with ratio of channel belt size to floodplain width, as noted by Leeder (1978), although in his experiments the variable quantity was floodplain width, whereas both channel belt width and depth are varied here. The effect illustrated here should be borne in mind in subsequent experiments where the channel belt size/floodplain width ratio is kept constant.

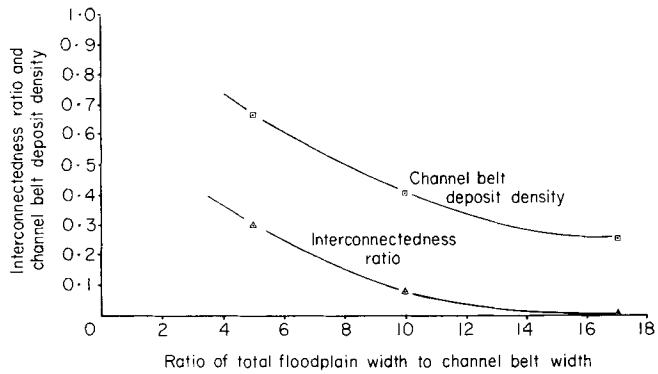

Fig. 3. Interconnectedness ratio and channel belt deposit density versus the ratio total floodplain width to channel belt width for set 1 experiments.

For set 2 experiments, Figs 4 and 5 show channel deposit density and interconnectedness to vary inversely with aggradation rate and avulsion periodicity, as deduced previously (Leeder, 1978). At the longer avulsion periodicities there is a marked increase in channel deposit density and interconnectedness ratio as channel-belt aggradation rate falls below about 0·01 m year^{-1}; above about 0·02 m year^{-1} these quantities are largely independent of aggradation rate. This tendency becomes progressively less obvious as avulsion periodicity decreases. As channel deposit density falls below about 0·4 interconnectedness ratio becomes very small, less than about 0·05 (as predicted theoretically by Allen, 1978). Allen's (1978) and Leeder's (1978) comparable values of channel deposit density and interconnectedness ratio are smaller than shown here for larger aggradation rates, probably because compaction of fines is neglected; otherwise their variation with aggradation rate show similar trends.

Let it be assumed that, all other variables being equal, a meandering stream has a lower channel belt aggradation rate and longer avulsion periodicity than a low sinuosity or braided stream, based on a much smaller sediment discharge in the meandering case (channel-belt size is the same in both cases). From Fig. 5 it appears that channel deposit density and interconnectedness could increase or decrease in moving from meandering to braided, depending on the precise nature of variation of avulsion rate and aggradation rate. It must be concluded that it would be extremely difficult to separate stream patterns based on interconnectedness ratio or channel deposit density in alluvial successions.

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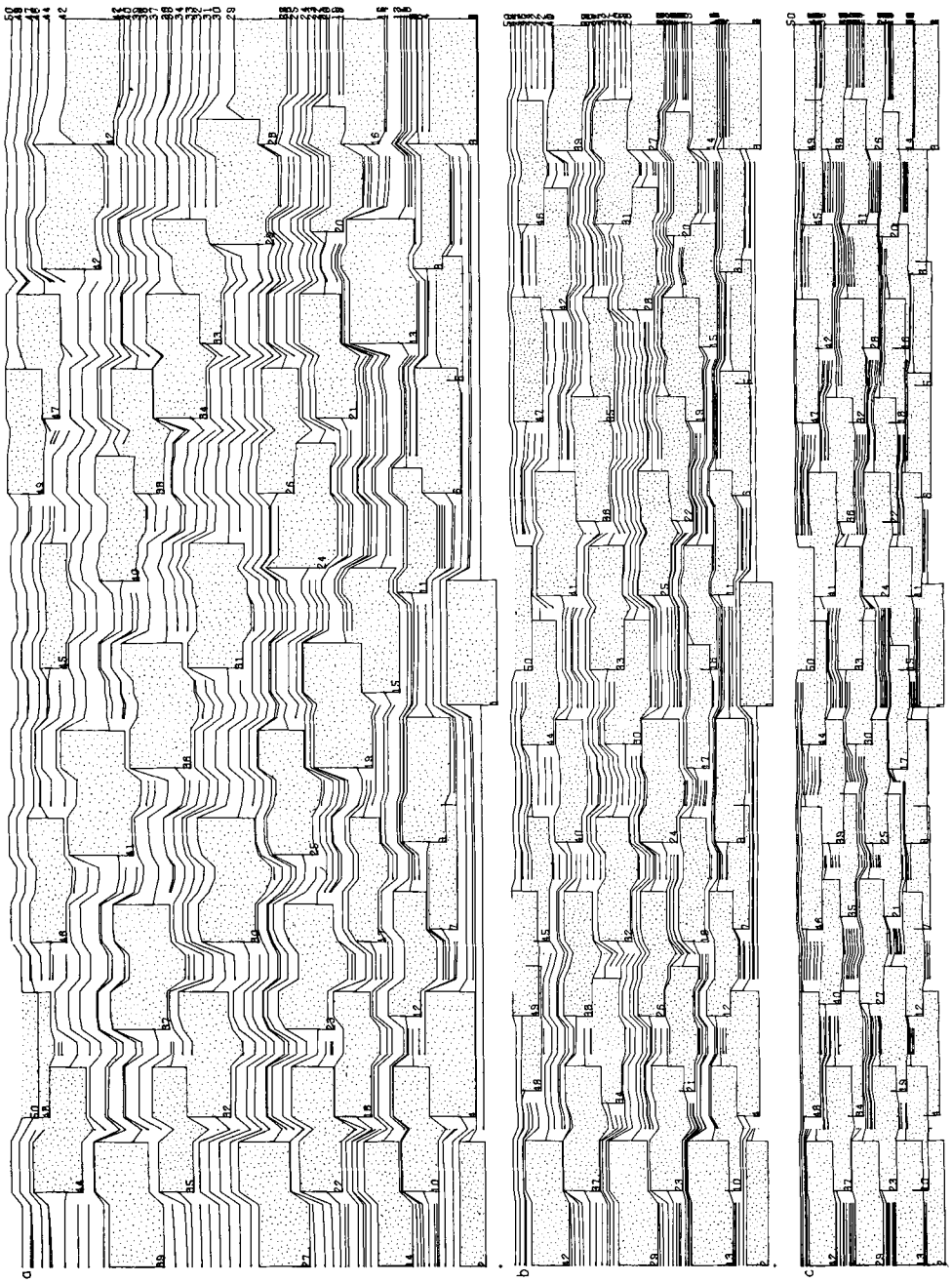

Fig. 4. Simulated cross-sections for selected set 2 experiments. Vertical exaggeration is 25. Mean avulsion period is 445 years in these examples; mean channel-belt aggradation rates are (a) 0·02 m year^{-1}, (b) 0·01 m year^{-1}, (c) 0·005 m year^{-1}.

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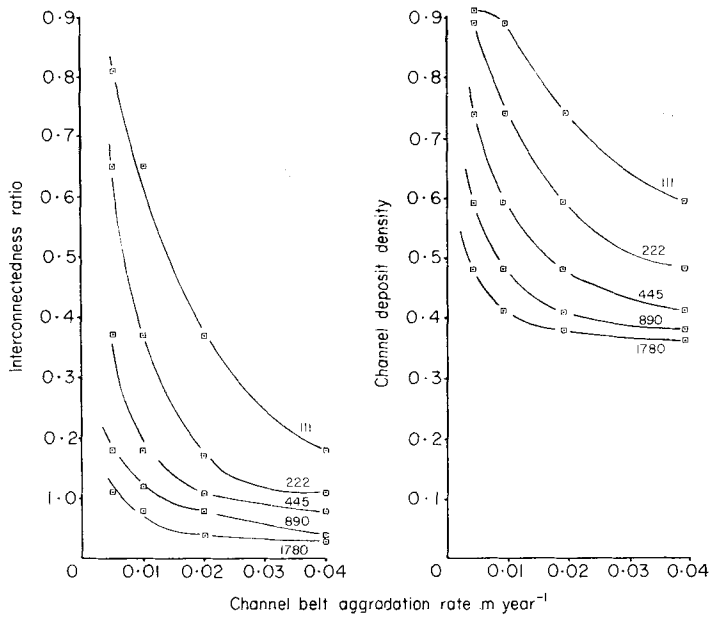

Fig. 5. Interconnectedness ratio and channel belt deposit density versus floodplain aggradation rate and avulsion periodicity for set 2 experiments. Numbers on curves represent mean avulsion periodicity.

For set 3 full graben runs (Fig. 6) the channel deposit density is unchanged by periodic fault movements along the alluvial basin margins. The interconnectedness ratio is slightly decreased by fault movement frequencies less than 445 years, compared to the case when no fault movement occurs at all.

For set 4 half graben runs (Fig. 7) channel deposit density averaged over the whole basin is again unchanged by periodic subsidence (but see next section). Infrequent fault movements give rise to interconnectedness ratios similar to the corresponding runs of set 2. Increasing frequency of fault movements at first causes a decrease in the interconnectedness ratio but at high frequency (<445 years) a very rapid increase is observed. As shown below, this latter trend is due to a channel deposit 'clustering' tendency adjacent to the actively downfaulting margin.

CHANNEL DEPOSIT CLUSTERING AND OFFLAP TENDENCIES

Histograms of the number of channel-belt occurrences in each reference vertical across the floodplains (e.g. Fig. 8) all show regular alternations of peaks (overlap-rich zones) and troughs (overlap-poor zones): For sets 1, 2, and 3 the mean spacing of the peaks is about 75% of channel-belt width, with approximately twice as many channel-belts comprising the peaks as in the troughs (this can be seen clearly in Figs 2, 4 and 6 without the aid of histograms). At high rates of floodplain tilting in the set 4 runs there is a marked clustering tendency of channel-belt deposits adjacent to the active

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basin margin, but the spacing and relative height of adjacent peaks and troughs only changes slightly (Figs 7 and 8). Only fine-grained alluvium may occur on the inactive basin side, and clustering causes marked increases in the interconnectedness ratios. The tendency is actually due to the frequent existence of a fault-induced transverse slope on the floodplain, causing avulsing channels to shift more frequently towards the fault zone than the passive zone. There are clear guidelines here for exploration drilling procedures in alluvial basins if channel belt width can be estimated and tectonic setting is known.

Progressive channel deposit offlap, giving rise to a diachronous alluvial sand-body was first proposed by Bersier (1958, Fig. 5) as a theoretical possibility for alluvial sequences. Behaviour of the Kosi River is now known to follow closely this model (Gole & Chitale, 1966). Parts of the Westwater Canyon Member, New Mexico, seem to show such an offlap pattern clearly (Campbell, 1976, Fig. 2). Channel deposit offlap tendencies are not obvious in our sets 1–3 because of our assumption that the avulsing channel will always move to the lowest position on the contemporary floodplain surface (but see Fig. 4a and b). In set 4 runs, however, there is an irregular tendency for periodic offlap away from the downfaulted zone during periods of quiescence, followed by channel avulsion back to the fault zone area after a down-faulting event (Fig. 7). Progressively younger offlapping channel belts do not necessarily follow each other sequentially.

THE NATURE OF CHANNEL BELT DEPOSITS

As the floodplain aggrades, the contemporary channel belt deposit must gradually thicken although channel depth remains constant. The amount of thickening will obviously depend directly upon channel-belt aggradation rate and residence time of the channel belt in a particular area of the floodplain (Leeder, 1978; Figs 4 and 9) From Fig. 9 it appears that the ratio between mean aggraded and original channel belt thickness will normally exceed unity. If it is assumed that avulsion frequency will increase with aggradation rate, however, there would appear to be a self-regulating upper limit to the ratio. This result is of some importance in the study of ancient channel deposits since it is commonly assumed (e.g. Allen, 1965; Leeder, 1973) that coarse member thickness may serve as a surrogate for original channel bankfull depth for meandering streams; such an assumption should obviously be treated with considerable scepticism.

Scour by the migrating thalwegs of aggrading channels within the channel belt would be expected to leave behind thick and repeatedly truncated lower bar and thalweg lag deposits (see also Bluck, 1971; Bridge, 1975). Possible ancient examples of aggraded channel belt deposits showing such complex cross-cutting relations between individual bar deposits (i.e. multistorey) are cited by Bersier (1958), Allen (1964), Conaghan & Jones (1975), Campbell (1976), Puigdefabregas & van Vliet (1978), and Nijman & Puigdefabregas (1978). In addition to the expected basal erosion surfaces within single channel belt deposits, however, a multistorey character will also be imposed by the superposition of distinct channel belts (see also Allen, 1978). It seems likely that these different orders of multistorey character would be difficult to distinguish in the field, particularly in areas of restricted exposure.

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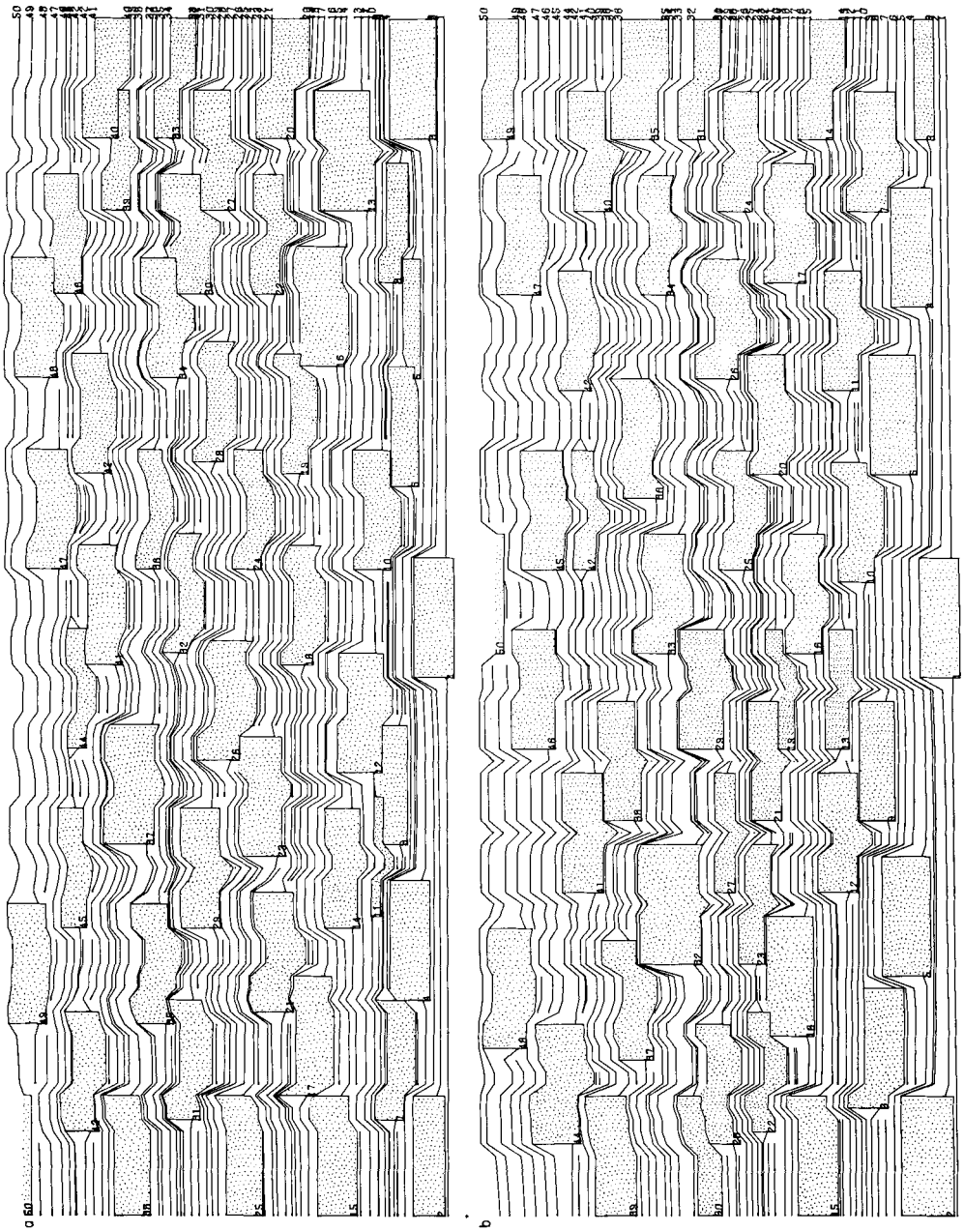

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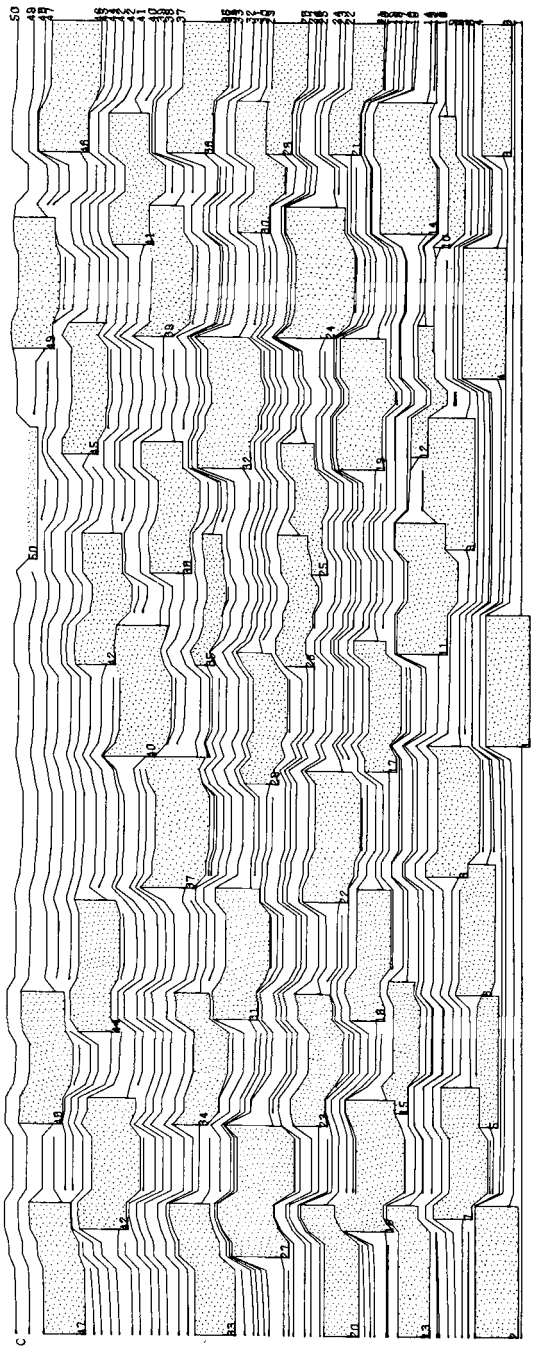

Fig. 6. Simulated cross-sections for selected set 3 (full graben) experiments. Vertical exaggeration is 12·5. Mean and standard deviation of relative movements at floodplain margins are 0·0 and 0·5 m respectively. Mean periods between tectonic events are (a) 890 years, (b) 445 years, (c) 89 years.

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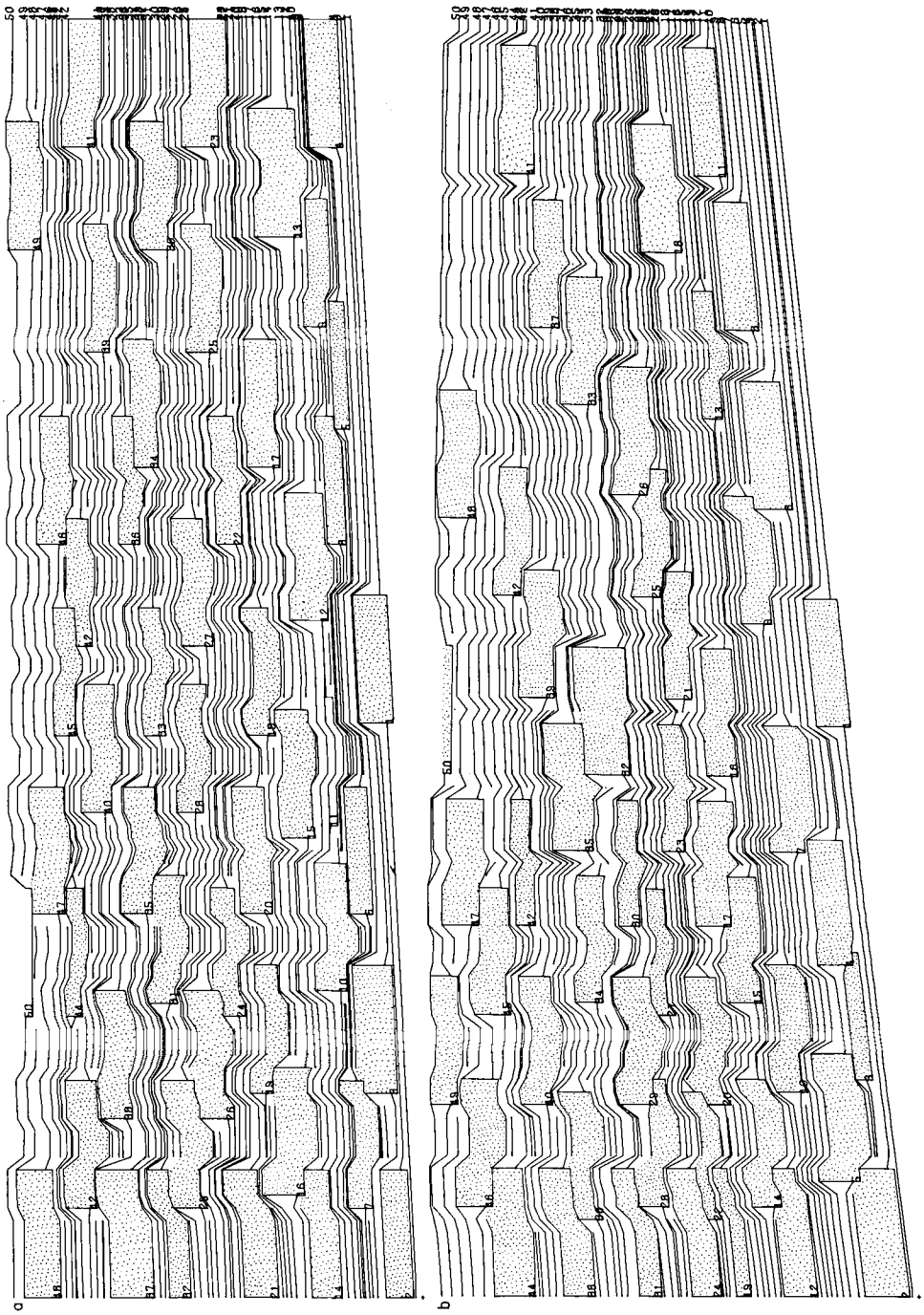

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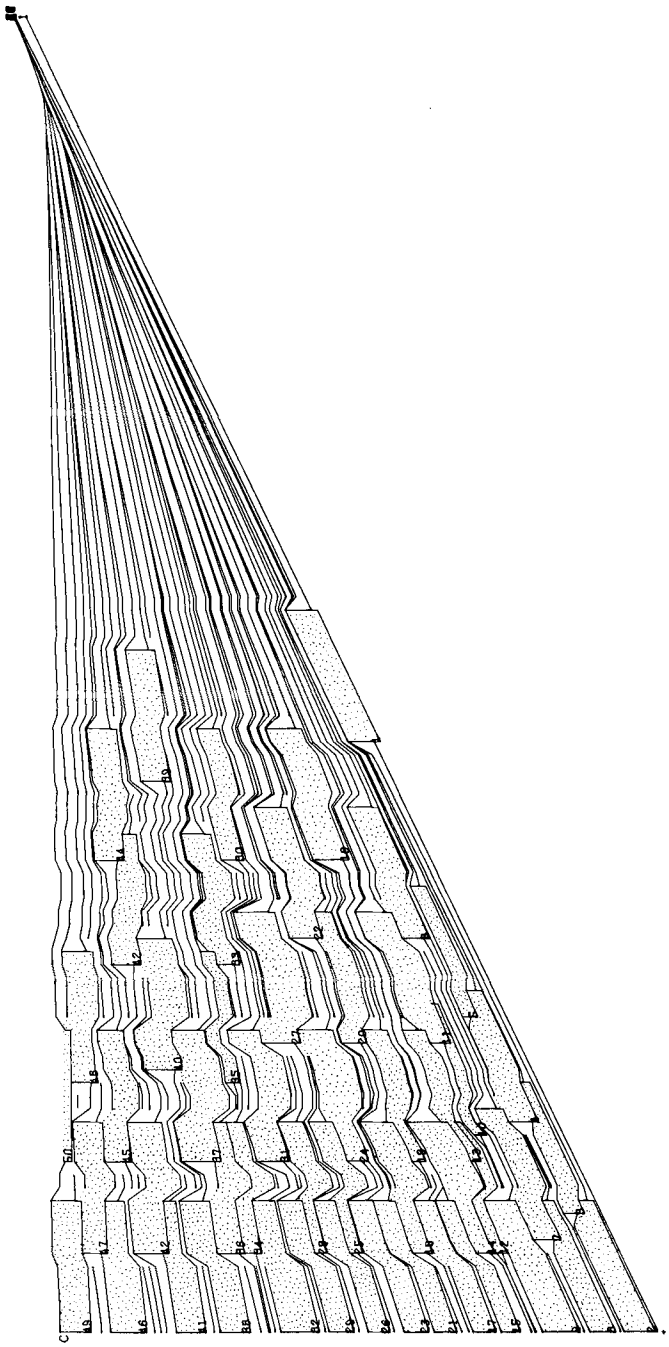

Fig. 7. Simulated cross-sections for selected set 4 (half graben) experiments. Vertical exaggeration is about 10. Mean and standard deviation of relative vertical movement at floodplain margins are 1·0 and 0·5 m respectively. Mean periods between tectonic events are (a) 890 years, (b) 445 years, (c) 89 years.

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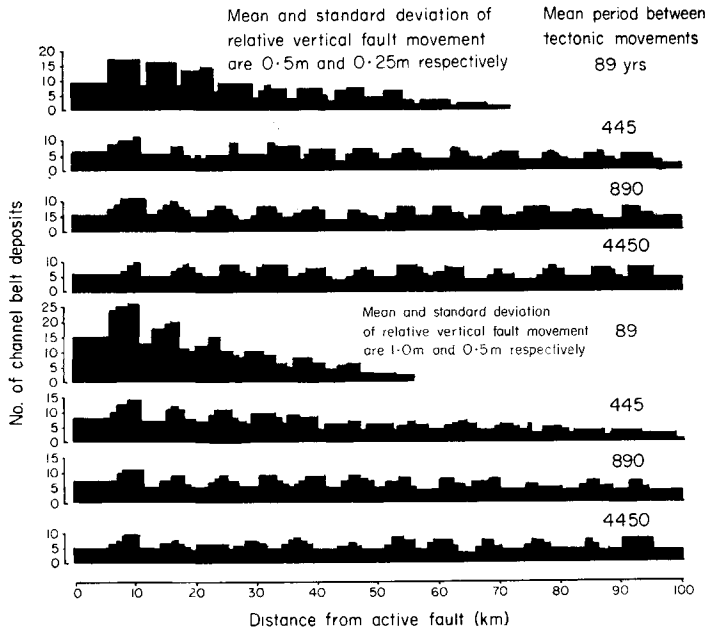

Fig. 8. Histograms to show the varying number of channel belt deposits encountered in floodplain cross-sections for the set 4 experiments. Active fault is on left side of floodplain.

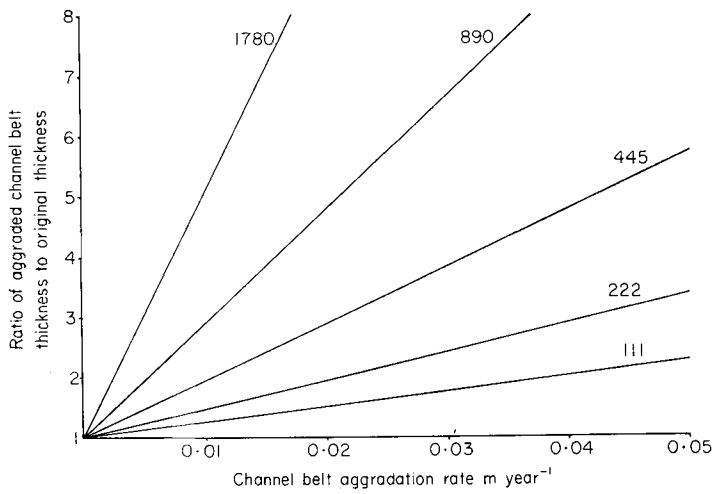

Fig. 9. Graph to show aggradational effects on channel belt deposit thickness. Numbers on lines show mean avulsion periodicity in years.

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CONCLUDING REMARKS

The composite model presented above cannot be tested directly on Recent or Quaternary alluvial sequences because relatively rapid climatic and sea level oscillations have conspired to produce subsurface sedimentary facies unrelated to contemporary patterns of river behaviour. The predicted successions, however, are in accordance with ancient two-dimensional sequences postulated to be of fluvial origin (Nanz, 1954; Friend & Moody-Stuart, 1972; Conaghan & Jones, 1975; Campbell, 1976, and many other authors). As well as helping to understand the origin of alluvial successions, the model may be used as a predictive tool in specific cases given some control from outcrop or borehole information. Development of the model has highlighted the need for much more empirical and theoretical research on processes affecting alluvial stratigraphy. Its design is flexible enough to accommodate future improvements and refinements to the existing submodels, or to include processes not yet considered. For example, improvements might be made in describing (a) the influence on local aggradation rates of complicated floodplain relief, or (b) the influence of aggradation, tectonism, and compaction on the probability of avulsion, or (c) time variation of channel-belt size and mean aggradation rate. The degradation process might also be considered.

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